CloakFi: Metasurface-Enabled Privacy Protection for Wi-Fi Integrated Sensing and Communication

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***Abstract*—Publicly available channel state information (CSI) in Wi-Fi has enabled a wide range of sensing applications and empowered integrated sensing and communication (ISAC). However, this same openness also raises serious privacy concerns. Existing defenses either neglect legitimate ISAC requirements or depend on costly, specialized hardware. To this end, we propose CloakFi, a plug-and-play, low-cost solution that leverages the channel manipulation capability of metasurfaces to safeguard privacy while preserving ISAC performance. The key idea is to assign each metasurface row two distinct beamforming vectors and randomly activate one at each time slot, introducing fluctuations that mask sensitive sensing information. Those vectors are jointly designed to remain nearly identical in the communication direction, ensuring stable, high-throughput transmission, while diverging in the sensing direction, thereby introducing significant fluctuations and obfuscating unauthorized observations. To further support legitimate sensing under such randomized configurations, we develop a time-domain masking and demasking method that associates CSI samples with their corresponding metasurface configurations and eliminates induced discrepancies, enabling accurate recovery of valid CSI. We implement CloakFi on commodity Wi-Fi devices, and extensive experiments demonstrate that it provides strong privacy protection without sacrificing communication or sensing performance for legitimate users.**

***Index Terms*—Integrated sensing and communications, Wi-Fi sensing, privacy protection, metasurface, wireless security**

I. Introduction

Wi-Fi signals, originally designed for communication, have recently been recognized as a powerful medium for wireless sensing. As these signals interact with the surrounding environment, reflecting off walls, scattering from objects, and diffracting around obstacles, they encode rich spatial and temporal information within their channel state information (CSI) [1], [2]. This insight has sparked growing interest in leveraging commodity Wi-Fi infrastructure for low-cost, device-free sensing. By analyzing CSI through signal processing and machine learning techniques, a wide range of sensing applications becomes possible, such as human activity recognition [3]–[5], respiration monitoring [6], [7], and trajectory tracking [8]. This sensing capability, when integrated into communication systems, forms the integrated sensing and communication (ISAC) [9]–[11], a paradigm that empowers wireless networks to both connect and sense.

While CSI enables powerful sensing capabilities, its accessibility also introduces subtle yet serious privacy risks. In Wi-Fi systems, CSI is not exclusive to legitimate users; nearby devices, even those not associated with the network, can passively extract it from over-the-air transmissions [12]. Imagine a person typing on a laptop in a quiet room. To the user, nothing seems exposed. Yet, nearby Wi-Fi signals reflect off their body and fingers, embedding behavioral cues into the wireless channel. Recent studies show that an attacker can intercept such signals and reconstruct human trajectories over time by analyzing CSI fluctuations [13], [14], and even infer sensitive actions like keystrokes [15]–[18]. These examples reveal a critical risk: in environments with Wi-Fi, private behaviors can be inferred by passive observers simply by analyzing the measured signals that propagate through space.

To address such risks, prior studies have explored various defenses aimed at mitigating privacy leakage caused by CSI. A straightforward approach perturbs the wireless channel by leveraging the ability of the transmitter (Tx), e.g., by varying transmit power to induce artificial CSI fluctuations [13], but this inevitably degrades communication quality. MIMOCrypt [19] achieves a better trade-off by randomizing precoding, yet it depends on multi-antenna hardware, making it impractical for low-cost, single-antenna Internet of Things (IoT) devices. Another line of work employs full-duplex jammers to inject interference, which can hinder unauthorized sensing but comes at high hardware cost and simultaneously disrupts sensing for nearby legitimate users [20]. The limitation of high cost has motivated the use of low-cost programmable metasurfaces, which can passively manipulate the propagation environment without requiring active transceivers or multi-antenna arrays [21], [22]. A notable example is IR-Shield [23] that leverages metasurfaces to randomize the channel and confuse eavesdroppers. However, its design only introduces limited perturbations by phase-randomizing selected metasurface regions and fails to fully exploit the beamforming potential of metasurfaces. More importantly, it overlooks the sensing and communication requirements of legitimate users. Consequently, existing solutions still fall short of providing a practical privacy-preserving design that can operate on low-cost IoT devices while simultaneously sustaining both communication and sensing performance.

To address this gap, we propose CloakFi, a metasurface-enabled privacy-aware ISAC system, as shown in Fig. 1; it fully exploits the beamforming capability of metasurfaces rather than perturbing only partial regions as in [23]. The key idea is to assign each metasurface row a pair of beamforming vectors. These vectors are carefully designed to produce signif-

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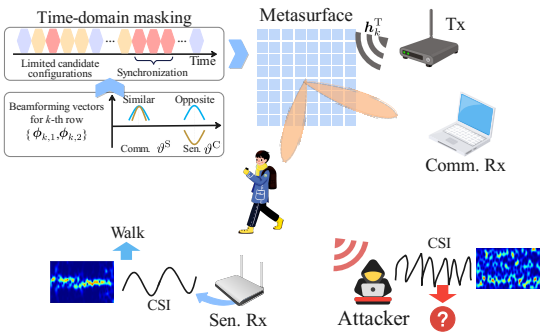


Fig. 1: CloakFi: metasurface is leveraged to achieve high-performance ISAC while preserving privacy.

icant discrepancies in the sensing direction while maintaining nearly identical gains in the communication direction. In this way, switching between the two vectors perturbs sensing signals without compromising communication. However, achieving this design is non-trivial since the directly formulated optimization problem is NP-hard, given that commercial metasurfaces typically support only 1-bit phase shifts. To tackle this challenge, we relax the discrete 1-bit phase shifts of the metasurface into continuous variables and incorporate a penalty term, thereby transforming the problem into a tractable optimization that jointly considers privacy-preserving perturbations and communication performance across all metasurface rows. To solve it, we develop an algorithm based on the block coordinate descent (BCD) method [24], which iteratively updates variable blocks and guarantees stable convergence to a feasible 1-bit solution.

Moreover, fully randomizing metasurface beamforming configurations can indeed disrupt eavesdroppers but also prevent legitimate receivers (Rx) from recovering valid sensing information, since the metasurface cannot actively cancel its own perturbations like full-duplex jammers [20]. To overcome this, we design a time-domain masking and de-masking mechanism. Instead of sampling from the entire configuration space, we construct a small subset of candidate configurations and randomly activate one per time slot. Restricting the choice to this subset allows the legitimate Rx to leverage channel coherence time, during which the underlying channel response remains nearly constant, to reliably estimate the effects introduced by different configurations. Concretely, to synchronize CSI samples with metasurface states, fixed configuration patterns are periodically inserted as anchors. By detecting these anchors, the Rx realizes synchronization with metasurface. It then estimates the relative gain caused by different configurations and compensates for them, thereby restoring a consistent CSI sequence and enabling legitimate sensing. In summary, we make the following major contributions:

- We present CloakFi, a metasurface-enabled ISAC system that ensures both privacy and high ISAC performance, even for low-cost IoT devices.
- We design a beamforming algorithm for metasurface, and it preserves stable communication while introducing controlled perturbations to obfuscate sensing information.
- We develop a time-domain masking and demasking method that preserves privacy while enabling legitimate sensing Rx to reliably extract information.
- We implement CloakFi on commodity devices, and extensive experiments confirm its effectiveness in safeguarding privacy without compromising ISAC performance.

The remainder of this paper is organized as follows. Section II provides background on Wi-Fi sensing and the attacker model. Section III introduces the Wi-Fi model with a metasurface and a feasibility study motivating the design of CloakFi. Section IV presents the detailed design of CloakFi. Sections V and VI describe the experiment setup and evaluation results. Section VII reviews related work, and the whole paper is concluded in Section VIII.

II. Preliminary and Attack Model

In this section, we introduce the basic knowledge on WiFi sensing and define the attack model.

A. Wi-Fi Sensing Basic

Wi-Fi sensing exploits the fact that wireless signals, while primarily used to carry data, also encode rich information about the surrounding environment. As signals propagate, they undergo multipath effects caused by reflection, scattering, and diffraction from objects and human bodies. These propagation changes are captured by the CSI, which can be estimated at the Rx by comparing the known pilot symbols embedded in transmitted packets with the received ones. Since CSI is highly sensitive to even subtle environmental dynamics, analyzing its temporal variations enables the detection and recognition of human activities, ranging from fine-grained motions such as keystrokes to larger-scale behaviors like walking.

B. Threat Model

Because pilot symbols are standardized and publicly known, any nearby device can overhear them. This means that an adversary does not need to be a legitimate Rx: simply by listening to ongoing transmissions, it can measure the corresponding CSI. In our setting, we consider a passive attacker situated within the environment, who leverages commodity Wi-Fi hardware or software-defined radios (SDRs) to capture packets and compute CSI that implicitly contains information about the target. By analyzing CSI, the adversary may infer user activities and even extract highly sensitive details, such as keystrokes or passwords. Meanwhile, we assume the adversary is endowed with the following abilities [19]:

- **Flexible placement**. The attacker can deploy its sniffing device anywhere in the monitored area. This allows it to select vantage points that maximize the chance of a successful attack.
- **Limited hardware**. In line with practical constraints of Wi-Fi cards or SDR platforms, the attacker typically controls no more than three antennas. For completeness, we will examine a stronger setting where multiple Wi-Fi devices are aggregated to form a larger antenna array, and verify that the attacker still fails.

- **Model access**. We consider the attacker to be well-informed and able to use the same pre-trained sensing model available to legitimate Rx. Thus, once CSI is collected, the adversary can directly apply the model to attempt behavior inference. This ensures that any observed protection effect arises from our design rather than from model deficiencies.

III. Metasurface and Feasibility Study

We first introduce the Wi-Fi with metasurface and then leverage simple experiments to motivate the design of CloakFi.

A. Wi-Fi with Metasurface

A metasurface is a two-dimensional programmable structure composed of many controllable reflecting units. By adjusting the voltage applied to each unit, the phase of the reflected signal can be reconfigured, thereby enabling dynamic control of the wireless channel. In this work, we leverage this feature to realize privacy protection. Specifically, a metasurface is deployed near the Tx, and a directional antenna steers the Tx's signal toward the metasurface, as shown in Fig. 1. For a metasurface with $K \times N$ elements, the passive beamforming matrix corresponding to the k-th row is defined as $\mathbf{\Phi}_k = \mathrm{diag}\{e^{j\phi_{k,1}}, \cdots, e^{j\phi_{k,N}}\} \in \mathbb{C}^{K\times N}$, where $\phi_{k,n}$ represents the phase of the (k,n)-th reflecting unit.

Let $\boldsymbol{h}_k^{\mathrm{T}} \in \mathbb{C}^{N\times 1}$ denote the wireless channel between the Tx and the k-th row of the metasurface. For the communication link, the channel from the k-th row of the metasurface to the communication Rx is denoted by $\boldsymbol{G}_k^{\mathrm{C}} \in \mathbb{C}^{M^{\mathrm{C}}\times N}$, and then the channel between the Tx and the communication Rx can be expressed as

$$\boldsymbol{h}^{\mathrm{Com}} = \sum\nolimits_{k=1}^{K} (\boldsymbol{G}_k^{\mathrm{C}} \mathbf{\Phi}_k \boldsymbol{h}_k^{\mathrm{T}}). \tag{1}$$

Given the presence of a strong line-of-sight (LoS) path between the metasurface and the communication Rx, the channel $\boldsymbol{G}_k^{\mathrm{C}}$ is primarily dominated by this LoS component with a_k^{C} being the path loss, θ^{C} being the angle of arrival (AoA), and ϑ^{C} being the angle of departure (AoD). Then, the channel can be reasonably approximated as: $\boldsymbol{G}_k^{\mathrm{C}} = a_k^{\mathrm{C}} \boldsymbol{\alpha}(\theta^{\mathrm{C}}) \boldsymbol{\alpha}^H(\vartheta^{\mathrm{C}})$, where $\boldsymbol{\alpha}(\cdot)$ is the steering vector. Meanwhile, by defining beamforming vector $\boldsymbol{\phi}_k \triangleq [e^{j\phi_{k,1}}, \cdots, e^{j\phi_{k,N}}]^T \in \mathbb{C}^{N\times 1}$ for k-the row of the metasurface and $\boldsymbol{h}_k^{\mathrm{C}} \triangleq (a_k^{\mathrm{C}} \boldsymbol{\alpha}^H(\vartheta^{\mathrm{C}}) \mathrm{diag}\{\boldsymbol{h}_k^{\mathrm{T}}\})^H \in \mathbb{C}^{N\times 1}$, the channel can be rewritten as

$$\boldsymbol{h}^{\mathrm{Com}} = \boldsymbol{\alpha}(\theta^{\mathrm{C}}) \left(\sum_{k=1}^{K} (\boldsymbol{h}_k^{\mathrm{C}})^H \boldsymbol{\phi}_k \right). \tag{2}$$

Since $\boldsymbol{\alpha}^H(\theta^{\mathrm{C}}) \boldsymbol{\alpha}(\theta^{\mathrm{C}}) = M^{\mathrm{C}}$, the communication signal-to-noise ratio (SNR) is

$$\mathrm{SNR}^{\mathrm{Com}} = M^{\mathrm{C}} || \sum_{k=1}^{K} (\boldsymbol{h}_k^{\mathrm{C}})^H \boldsymbol{\phi}_k ||^2 P^{\mathrm{T}} / \sigma^2, \tag{3}$$

where P^{T} denotes the transmit power at the Tx, and σ^2 is the power of the complex Gaussian noise at the Rx.

Similarly, for the sensing link, the channel from the k-th row of the metasurface to the sensing Rx is denoted by $\boldsymbol{G}_k^{\mathrm{S}} \in \mathbb{C}^{M^{\mathrm{C}}\times N}$. It mainly contains two parts: the dynamic path $\boldsymbol{G}_k^{\mathrm{S,S}}$ related to the sensing target and the static part $\boldsymbol{G}_k^{\mathrm{S,O}}$ consisting of other paths, i.e., $\boldsymbol{G}_k^{\mathrm{S}} = \boldsymbol{G}_k^{\mathrm{S,S}} + \boldsymbol{G}_k^{\mathrm{S,O}}$. Moreover, the former can be expressed as $\boldsymbol{G}_k^{\mathrm{S,S}} = a_k^{\mathrm{S}} \boldsymbol{\alpha}(\theta^{\mathrm{S}}) \boldsymbol{\alpha}^H(\vartheta^{\mathrm{S}})$, where a_k^{S} is the path loss, θ^{S} is the AoA, and ϑ^{S} is the AoD. Thus, the CSI measured at the sensing Rx is

$$\boldsymbol{h}^{\mathrm{Sen}} = \sum_{k=1}^{K} ((\boldsymbol{G}_k^{\mathrm{S,S}} + \boldsymbol{G}_k^{\mathrm{S,O}}) \mathbf{\Phi}_k \boldsymbol{h}_k^{\mathrm{T}}). \tag{4}$$

By defining $\boldsymbol{h}_k^{\mathrm{S}} \triangleq (\boldsymbol{\alpha}(\vartheta^{\mathrm{S}})^H \mathrm{diag}\{\boldsymbol{h}_k^{\mathrm{T}}\})^H$, the transmit power towards the direction of the sensing target (i.e., ϑ^{S}) for the k-th metasurface row can be derived as

$$P_k^{\mathrm{Sen}} = ||(\boldsymbol{h}_k^{\mathrm{S}})^H \boldsymbol{\phi}_k||^2 P^{\mathrm{T}}. \tag{5}$$

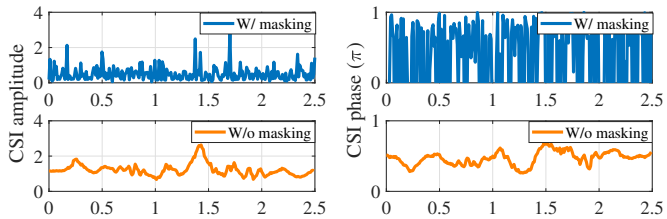
From equations (4) and (5), we observe that adjusting $\boldsymbol{\phi}_k$ (or equivalently $\mathbf{\Phi}_k$) can cause variations in both the CSI and the received power at the sensing Rx. Leveraging this property, we propose to equip each row of the metasurface with two predefined passive beamforming vectors, denoted as $\boldsymbol{\phi}_{k,1}$ and $\boldsymbol{\phi}_{k,2}$. We generate N^{R} different metasurface configurations by randomly selecting one beamforming vector for each metasurface row. Switching between those configurations in a randomized manner can intentionally induce additional randomness into the wireless channel. In this paper, the objectives can be summarized as follows:

- Maintain a reliable and high-rate communication link between the Tx and the communication Rx.
- Prevent privacy leakage of the target through CSI, regardless of an eavesdropper's position.
- Allow the legitimate sensing Rx to recover target-relevant features, thereby ensuring high-quality sensing.

B. Feasibility Study and Motivation

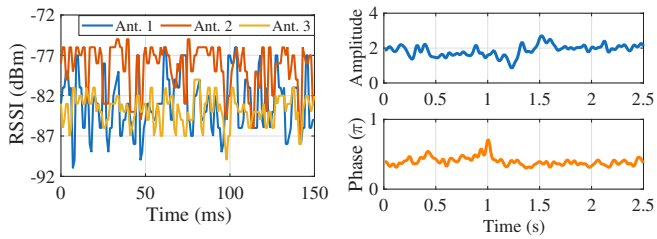
To motivate the design of CloakFi, we carry out a simple feasibility study. In particular, we employ three Intel 5300 WiFi NICs to build the prototype system. The Tx (the first NIC) continuously transmits data packets using one antenna, while an 8×16 metasurface is deployed to introduce perturbations into the wireless channel. The sensing Rx, also based on an Intel 5300 NIC, records CSI, and a third NIC serves as the communication Rx. During the experiment, a human subject repeatedly executes a "slide" gesture, and the detailed experiment setup and layout are provided in Section V. To deliberately amplify CSI randomness introduced by metasurface configurations, we first design the initial beamforming vector $\boldsymbol{\phi}_{k,1}$ by maximizing the sensing power P_k^{Sen}. The second vector $\boldsymbol{\phi}_{k,2}$ is then obtained by applying an additional π phase shift to each element of $\boldsymbol{\phi}_{k,1}$, i.e., $\boldsymbol{\phi}_{k,2} = -\boldsymbol{\phi}_{k,1}$, ensuring the largest possible discrepancy along the sensing direction. Moreover, we only construct four distinct metasurface configurations, i.e., $N^{\mathrm{R}} = 4$.

Privacy protection. Fig. 2 presents the CSI measured at the sensing Rx. Without the metasurface, both amplitude and phase traces show clear and stable structures, which reveal gesture-related dynamics. Once the metasurface is applied,



(a) Amplitude. (b) Phase.

Fig. 2: (a) Amplitude and (b) phase of the CSI with and without masking using metasurface.



(a) RSSI. (b) CSI after demasking.

Fig. 3: (a) The communication performance and (b) the CSI after demasking.

however, the CSI curves become irregular and noisy, thereby concealing sensing cues and reducing the risk of information leakage. *This result indicates that, when two beamforming vectors assigned to each metasurface row are designed to create strong discrepancy in the sensing direction, a small number of metasurface configurations (e.g., four) are sufficient for introducing notable perturbations and safeguarding privacy.*

Communication performance. When continuously and randomly activating a metasurface configuration, the communication Rx experiences pronounced variations in received signal strength indicator (RSSI), reflecting poor link stability, as illustrated in Fig. 3(a). The primary reason is that the current beamforming vectors are designed without considering communication demands. Consequently, random switching among these vectors leads to severe SNR fluctuations at the communication end and may even disrupt the link. *This observation highlights the necessity of a revised passive metasurface beamforming design, one that maintains reliable communication performance while still producing adequate perturbations in the sensing direction.*

Sensing of legitimate Rx. For accurate sensing at the legitimate Rx, it is crucial to suppress the interference caused by the metasurface. Ideally, one might consider estimating the individual channels between each metasurface row and the Rx. Yet, this is infeasible in practice because the metasurface operates passively and cannot provide per-row channel measurements, unlike the full-duplex transceivers in [20]. Fortunately, since the CSI corresponds to only a limited number of metasurface configurations, once the received CSI samples are correctly mapped to their respective configurations and the configuration-dependent variations (arising from metasurface beamforming) are compensated, a reliable CSI sequence can be reconstructed for sensing purposes. The CSI recovered with this procedure is depicted in Fig. 3(b). Compared with the raw CSI shown in Fig. 2, the reconstructed signal still preserves recognizable patterns that remain informative for sensing tasks. *Nonetheless, for this reconstruction approach to be effective, two critical challenges must be addressed: (1) synchronizing the Rx with the metasurface to guarantee that each CSI sample is accurately aligned with its configuration, and (2) compensating the configuration-dependent discrepancies so that their adverse influence on sensing is removed.* To tackle these issues, we will introduce a time-domain masking and demasking method.

IV. The Design of CloakFi

This section presents the workflow of CloakFi. The Tx first receives access requests from both the legitimate communication Rx and the sensing Rx. Upon receiving the requests, the Tx executes Algorithm 1 to determine the metasurface beamforming vectors. Simultaneously, a digital key (used for generating the metasurface configuration variations over time) is securely exchanged between the Tx and the sensing Rx to support encrypted sensing. Once the initialization phase is complete, the Tx configures the metasurface with the optimized beamforming vectors and begins continuous packet transmission to enable communication and sensing. Upon receiving the packets, the sensing Rx measures the CSI, applies the shared key to recover the sensing information, and performs sensing using a designated sensing algorithm. CloakFi involves two key components: a beamforming design that jointly considers the privacy protection and communication performance in Section IV-A, and a time-domain masking and demasking method that enables the legitimate sensing Rx to achieve high-performance sensing in Section IV-B.

A. Metasurface Beamforming Design

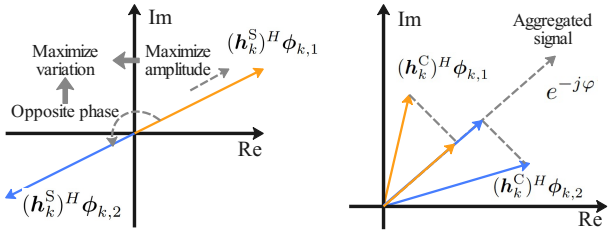
In this section, we first formulate an optimization problem for metasurface passive beamforming design and propose a BCD-based algorithm to solve it.

1) Design Objective: As indicated in Section III-B, an advanced beamforming scheme is needed, and the two beamforming vectors ($\boldsymbol{\phi}_{k,1}$ and $\boldsymbol{\phi}_{k,2}$) for each metasurface row should fulfill the following objectives:

- **Privacy preservation:** As illustrated in Fig. 4(a), both vectors should steer power toward the sensing direction to induce pronounced CSI fluctuations. This is achieved by maximizing the aggregate sensing gain: $\max ||(\boldsymbol{h}_k^{\mathrm{S}})^H \boldsymbol{\phi}_{k,1}||^2 + ||(\boldsymbol{h}_k^{\mathrm{S}})^H \boldsymbol{\phi}_{k,2}||^2$. In addition, their signals should have comparable amplitudes but nearly opposite phases, which can be approximated by minimizing $\min ||(\boldsymbol{h}_k^{\mathrm{S}})^H \boldsymbol{\phi}_{k,1} + (\boldsymbol{h}_k^{\mathrm{S}})^H \boldsymbol{\phi}_{k,2}||^2$.
- **Communication performance:** The aggregated channel across all rows must remain stable to ensure high data rates. Since each row randomly selects between two vectors, we optimize the worst-case SNR:

$$\max \min_{\forall x_k} \frac{M^{\mathrm{C}} \left\| \sum_{k=1}^{K} x_k (\boldsymbol{h}_k^{\mathrm{C}})^H \boldsymbol{\phi}_{k,1} + (1-x_k)(\boldsymbol{h}_k^{\mathrm{C}})^H \boldsymbol{\phi}_{k,2} \right\|^2 P^{\mathrm{T}}}{\sigma^2}, \tag{6}$$

where $x_k \in \{0, 1\}$ denotes the vector choice for the k-th row.



(a) Privacy preservation (b) Communication performance

Fig. 4: Two objectives for problem formulation.

The two objectives above jointly enforce effective sensing obfuscation while sustaining reliable communication. Nevertheless, the worst-case formulation in (6) is highly intractable, since it requires optimizing over all possible row-wise vector selections. To derive a more tractable surrogate, we note that the original objective essentially maximizes the squared magnitude of a sum of complex signals. This value is maximized when individual components are both large in amplitude and well phase-aligned. Motivated by this intuition, we define the reference phase of the aggregated signal as $\varphi = \angle\left(\sum_{k=1}^{K}(\boldsymbol{h}_k^{\mathrm{C}})^H\boldsymbol{\phi}_k\right)$. As illustrated in Fig. 4(b), we then rotate each candidate beamformed signal by $e^{-j\varphi}$ such that their alignment with the reference can be measured by their real part. Finally, we maximize the worst-case projection across the two candidate vectors:

$$\max \min_{i\in\{1,2\}} \mathcal{R}\{(\boldsymbol{h}_k^{\mathrm{C}})^H\boldsymbol{\phi}_{k,i}e^{-j\varphi}\}. \tag{7}$$

2) Problem Formulation: Since practical metasurfaces typically offer only 1-bit phase control, i.e., $\boldsymbol{\phi}k,i[n] \in -1,1$, the design task reduces to an integer programming problem, which is generally NP-hard. To circumvent this difficulty, we relax the binary constraint by allowing $\boldsymbol{\phi}k,i[n]$ to take continuous values in $[-1,1]$ and introduce a penalty to steer the solution toward feasible binaries. Specifically, we append a term $\rho\big((\boldsymbol{\phi}_{k,i}[n])^2 - 1\big)$ to the objective, where ρ is a tunable penalty factor. This term vanishes when $\boldsymbol{\phi}_{k,i}[n] = \pm 1$ and becomes negative otherwise, thereby suppressing invalid values and biasing the optimization toward $\{-1,1\}$. Then, the problem can be reformulated as

$$\begin{aligned}
\max_{\boldsymbol{\phi}_{k,i},\varphi} \quad & \sum_{k=1}^{K}\Bigg(\omega_1\left\|(\boldsymbol{h}_k^{\mathrm{S}})^H\boldsymbol{\phi}_{k,1}\right\|^2 + \omega_1\left\|(\boldsymbol{h}_k^{\mathrm{S}})^H\boldsymbol{\phi}_{k,2}\right\|^2 \\
& -\omega_2\left\|(\boldsymbol{h}_k^{\mathrm{S}})^H\boldsymbol{\phi}_{k,1} + (\boldsymbol{h}_k^{\mathrm{S}})^H\boldsymbol{\phi}_{k,2}\right\|^2 \\
& +\omega_3 \min_{i\in\{1,2\}} \mathcal{R}\{(\boldsymbol{h}_k^{\mathrm{C}})^H\boldsymbol{\phi}_{k,i}e^{-j\varphi}\}\Bigg) \\
& + \sum_{k,i,n}\rho((\boldsymbol{\phi}_{k,i}[n])^2 - 1), \qquad (8a)\\
\text{s.t.} \quad & -1 \le \boldsymbol{\phi}_{k,i}[n] \le 1,\ \forall k,i,n, \qquad (8b)
\end{aligned}$$

where ω_1, ω_2, and ω_3 are weighting factors.

3) Beamforming Design: To tackle problem (8), we employ a BCD scheme, which partitions the variables into blocks and updates them in turn. The procedure follows three steps: 1) fixing all other variables, each entry of the beamforming vector $\boldsymbol{\phi}_{k,1}$ is updated sequentially; 2) the same process is applied to $\boldsymbol{\phi}_{k,2}$; 3) given both vectors, the communication-phase variable φ is subsequently refined.

Algorithm 1: The beamforming design algorithm.

Define the tolerance of accuracy δ. Initialize the algorithm with a feasible point. Set the maximum iteration number $L_{\max}$ and the penalty factor ρ;
repeat
  Set iteration number $l = 0$;
  repeat
    Update $\boldsymbol{\phi}_{k,1}[n]$, $\forall n,k$ according to Step 1;
    Update $\boldsymbol{\phi}_{k,2}[n]$, $\forall n,k$ according to Step 2;
    Update φ according to Step 3;
    Update the iteration number: $l \leftarrow l+1$;
  until *The decrease of the objective function is less than δ or $l \ge L_{\max}$*
  Update $\rho \leftarrow 2\rho$;
until $\sum_{k,i,n}||\boldsymbol{\phi}_{k,i}[n]| - 1| \le \delta$.

Focusing on Step 1, we optimize each element $\boldsymbol{\phi}k,1[n]$ for all k,n. After direct mathematical derivation, the subproblem reduces to the following quadratic form:

$$\begin{aligned}
\max_{\boldsymbol{\phi}_{k,1}[n]\in\mathbb{R}} \quad & \beta_{k,n,1}(\boldsymbol{\phi}_{k,1}[n])^2 + \beta_{k,n,2}\boldsymbol{\phi}_{k,1}[n] \\
& + \min\{\beta_{k,n,3}\boldsymbol{\phi}_{k,1}[n], \beta_{k,n,4}\}, \qquad (9a)\\
\text{s.t.} \quad & -1 \le \boldsymbol{\phi}_{k,1}[n] \le 1, \qquad (9b)
\end{aligned}$$

where $\beta_{k,n,1}$, $\beta_{k,n,2}$, $\beta_{k,n,3}$, and $\beta_{k,n,4}$ denote parameters whose explicit expressions are omitted for brevity. This quadratic problem can be efficiently solved using standard optimization techniques.

In Step 2, we sequentially optimize $\boldsymbol{\phi}_{k,2}[n]$ for all k and n. Since this step closely resembles Step 1, we omit the detailed derivations. In Step 3, we optimize the phase variable φ, and the corresponding optimization problem is formulated as:

$$\max_{\varphi} \sum_{k=1}^{K}\min_{i\in\{1,2\}} \mathcal{R}\{(\boldsymbol{h}_k^{\mathrm{C}})^H\boldsymbol{\phi}_{k,i}e^{-j\varphi}\}. \tag{10}$$

The main challenge here arises from the non-smoothness of the objective function due to the $\min$ operator. To address this, we first identify the switching points for each k:

$$\varphi_k^{\mathrm{sw}} = -\angle((\boldsymbol{h}_k^{\mathrm{C}})^H\boldsymbol{\phi}_{k,1} - (\boldsymbol{h}_k^{\mathrm{C}})^H\boldsymbol{\phi}_{k,2}) + \frac{\pi}{2} + i\pi,\ i\in\mathbb{Z}, \tag{11}$$

at which $\mathcal{R}\{(\boldsymbol{h}_k^{\mathrm{C}})^H\boldsymbol{\phi}_{k,1}e^{-j\varphi}\} = \mathcal{R}\{(\boldsymbol{h}_k^{\mathrm{C}})^H\boldsymbol{\phi}_{k,2}e^{-j\varphi}\}$. Restricting φ_k^{sw} to $[0,2\pi]$ yields at most $2K$ distinct breakpoints, dividing the domain into subintervals where the minimizing index $i_k \in \{1,2\}$ remains fixed. Within each subinterval, the objective reduces to a smooth function, i.e., $\max\ \mathcal{R}\{\sum_{k=1}^{K}(\boldsymbol{h}_k^{\mathrm{C}})^H\boldsymbol{\phi}_{k,i_k}e^{-j\varphi}\}$. We solve this problem for each subinterval and then determine the global optimum by selecting the best among these candidates.

Until now, we have obtained the overall algorithm, as shown in Algorithm 1. It follows a double-loop structure: the

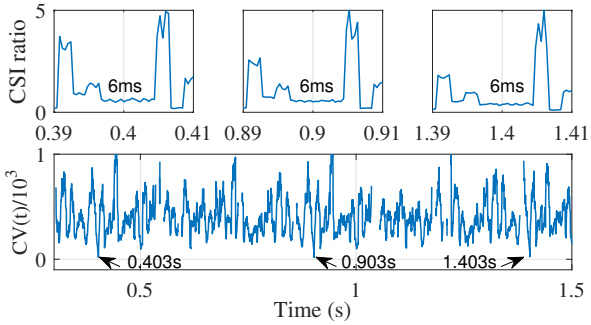


Fig. 5: Time synchronization between the Rx and the metasurface with T^{MTS} being 2 ms and T^{sync} being 0.5 s.

outer loop updates the penalty factor ρ, while the inner loop applies the BCD method for iterative optimization. Regarding convergence, each subproblem in Algorithm 1 is solved exactly and optimally within the BCD framework. Therefore, according to [25], Algorithm 1 is guaranteed to converge to a Karush–Kuhn–Tucker (KKT) point of problem (8).

B. Time-Domain Masking and Demasking

Up to this point, we have introduced a beamforming design for the metasurface. By randomly selecting a beamforming vector per row, we generate N^{R} candidate configurations, where N is kept small to ensure tractability. In each interval T^{MTS}, one configuration is randomly activated, thereby inducing temporal variations that mask sensitive features (i.e., Fig. 2) and enhance privacy protection. The subsequent challenge lies in ensuring that the legitimate sensing Rx can still recover reliable target information. As outlined in Section III-B, this requires the Rx to remain synchronized with the metasurface to identify the configuration at each time slot, while simultaneously unifying the CSI across configurations to remove perturbation and obtain meaningful sensing features.

1) Synchronization: To ensure correct mapping between received CSI samples and the metasurface configuration, precise synchronization between the metasurface and the Rx is required. Directly inferring the metasurface state from raw CSI dynamics is unreliable; instead, we introduce a synchronization mechanism by inserting a predefined beamforming vector that remains fixed for a short interval (e.g., $3T^{\mathrm{MTS}}$) within every T^{sync} period. This static segment creates a detectable marker that the Rx can exploit as a synchronization reference.

On the Rx side, the incoming CSI is compared with samples from the previous $3T^{\mathrm{MTS}}$ window to identify these stable segments. To mitigate the influence of time-varying phase offset, we adopt a CSI ratio [26] by dividing the CSI values between antenna pairs, thus generating time-series signals that can sensitively reflect the CSI variations. For a three-antenna WiFi NIC, this yields three ratio streams: antenna 1 over 2, 2 over 3, and 3 over 1, denoted as $h_m^{\mathrm{S,R}}(t)$. To determine whether the CSI is in a static state, we apply the coefficient of variation (CV), which measures relative signal fluctuation independent of absolute amplitude. For each subcarrier f and antenna ratio stream m, it is defined as:

$$\mathrm{CV}_{f,m}(t) = \frac{\mathrm{SD}_{\tau\in[t-3T^{\mathrm{MTS}},t]}\{h_m^{\mathrm{S,R}}(\tau)\}}{|\mathrm{Mean}_{\tau\in[t-3T^{\mathrm{MTS}},t]}\{h_m^{\mathrm{S,R}}(\tau)\}|}, \tag{12}$$

where $\mathrm{SD}\{\cdot\}$ and $\mathrm{Mean}\{\cdot\}$ represent the standard deviation and mean over time, respectively. To further suppress random noise, we further aggregate the CV values over all subcarriers and ratio streams:

$$\overline{\mathrm{CV}}(t) = \sum\nolimits_f \sum\nolimits_m \mathrm{CV}_{f,m}(t). \tag{13}$$

Because the static segment appears once in every T^{sync} interval and persists for only a few milliseconds, each interval contains exactly one distinct minimum in $\overline{\mathrm{CV}}(t)$. This minimum corresponds to the end of the segment and serves as a unique synchronization point. As illustrated in Fig. 5, where the interval between static segments is set to 0.5 seconds, each window exhibits a clear global minimum precisely aligned with the synchronization marker. To further improve accuracy, the Rx can apply linear least-squares estimation over multiple synchronization marker.

2) Unification: Once synchronization is established, the received CSI samples can be consistently aligned with the N^{R} candidate metasurface configurations through the shared key, denoted as $h_n^{\mathrm{Sen}}(t), n = 1, \cdots, N^{\mathrm{R}}$. According to the sensing channel model (4), different metasurface configurations yield varying sensing gains. To effectively suppress artificial perturbations, these gains must be compensated. Since the gains are not directly obtained at the Rx, we need to estimate them.

The estimation proceeds in two stages. First, we remove the contribution of static and target-independent paths. Specifically, the temporal mean of each sequence $h_n^{\mathrm{Sen}}(t)$ is subtracted from its raw samples:

$$\bar{h}_n^{\mathrm{Sen}}(t) = h_n^{\mathrm{Sen}}(t) - \underset{\text{over } t}{\mathrm{Mean}}\{h_n^{\mathrm{Sen}}(t)\},\ \forall n. \tag{14}$$

This operation highlights the dynamic component that carries sensing information. Second, since absolute gain recovery is infeasible, we instead focus on relative gains. One configuration (e.g., the first) is fixed as a reference, and all others are normalized with respect to it. Leveraging the fact that CSI remains nearly constant within a short interval, we traverse the CSI sequence to identify adjacent packets whose inter-packet intervals are below a predefined threshold and that belong to different configurations. By dividing their values, we estimate the relative gain w_{n_1,n_2} between the n_1-th and n_2-th configurations. Multiple estimates are aggregated, and their mean $\bar{w}_{n_1,n_2}$ is used to reduce noise effects. Collecting these estimates forms a relative gain matrix $\mathbf{W} = [\bar{w}_{n_1,n_2}] \in \mathbb{R}^{N^R\times N^R}$, which has unit diagonal entries.[1] Denoting the relative gain of the n-th configuration as g_n, the following least-squares problem is formulated:

$$\min_{g_n} \quad \sum\nolimits_{n_1=1}^{N^R} \sum\nolimits_{n_2=n_1+1}^{N^R} \left|g_{n_1} - \bar{w}_{n_1,n_2} g_{n_2}\right|^2, \tag{15a}$$

[1]The successful acquisition of the matrix is attributed to our deliberate restriction to a small configuration set.

$$\text{s.t.} \quad g_1 = 1. \tag{15b}$$

The goal is to find a set of gains $\{g_n\}$ that best fit the relative gain matrix $\mathbf{W}$, with the reference configuration fixed to $g_1 = 1$. This problem is quadratic and can be efficiently solved via convex optimization. After obtaining g_n, we normalize the CSI sequences as

$$\hat{h}_n^{\text{Sen}}(t) = \bar{h}_n^{\text{Sen}}(t)/g_n, \forall n. \tag{16}$$

The normalized CSI from all configurations are then reassembled in chronological order. To further suppress residual noise, a low-pass filter is applied. As shown in Fig. 3(b), the filtered CSI exhibits distinct temporal structures that preserve informative variations. Finally, these CSI sequences across antennas and subcarriers are fed into sensing algorithms for downstream sensing tasks.

V. Prototype and Experiment Setup

This section provides an overview of CloakFi's implementation and the experiment setup used for evaluation.

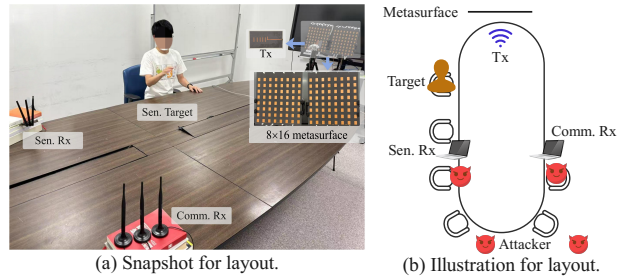
A. Implementation

Metasurface Prototype. Following [27], we design an 8×8 metasurface prototype with 64 reconfigurable elements. Each element employs a MADP-000907-14020x PIN diode to achieve 1-bit phase modulation, where bias voltages of 0 V and 1.35 V toggle the reflection phase between 0 and π. The element structure follows a conventional stacked metal–dielectric design optimized for 5.22 GHz. For control, an FPGA (ALINX AXU2CGB) manages the 64 units via serial-in, parallel-out shift registers (SN74HC595), which expand the limited I/O pins into 8-bit parallel control lines. This design allows efficient sequential loading of configuration bits and simultaneous phase updates across all elements.

System Implementation. CloakFi is composed of a Tx, a sensing Rx, a communication Rx, and a metasurface controlled by an FPGA. Both the Tx and Rxs are implemented on mini PCs equipped with Intel 5300 NICs. To emulate a low-cost IoT device, the Tx operates with a single transmit antenna, whereas each Rx is configured with three antennas to enable spatial diversity. The metasurface itself is formed by two 8×8 panels combined into a single 8×16 array, as illustrated in Fig. 6. On the software side, the FPGA is programmed in Verilog to implement both the metasurface beamforming vectors derived from Algorithm 1 and the time-domain masking method. For sensing, the Rx collects CSI using the PicoScene [28]. The demasking method is executed in MATLAB, followed by sensing algorithm development in Python. Here, we adopt the gesture classification model in SignFi [29] for sensing. Moreover, model training and evaluation are carried out on a workstation equipped with an NVIDIA RTX A5000 GPU.

B. Experiment Setup

We first conduct a study to validate the beamforming design in Section IV-A. The channel $\boldsymbol{h}_k^{\mathrm{T}}$ between the Tx and each metasurface element is estimated from the geometric distance between the transmit antenna and the corresponding element.



(a) Snapshot for layout. (b) Illustration for layout.

Fig. 6: Prototype and experiment setup.

Once verified, the derived beamforming vectors are applied to evaluate the overall system performance. Experiments are carried out in a standard meeting room environment, as illustrated in Fig. 6. All devices operate at 5.22 GHz with a 20 MHz bandwidth. The Tx continuously sends packets at approximately 500 Hz, while the metasurface updates its beamforming state every 2 ms. The sensing target is positioned at 50° and the communication Rx at -20° relative to the metasurface. By default, the attacker is placed at a distant location to emulate eavesdropping; additional tests consider three other attacker positions to assess robustness. We recruit six volunteers (four male, two female) who each perform nine gestures: push-pull (PP), slide (SL), up-down (UD), clap (CL), wave (WA), draw circle (DC), draw square (DS), draw zigzag (DZ), and idle (IS). Under the default setup, each gesture is repeated 50 times per participant, and the resulting dataset is split into training and testing with a 7:3 ratio. For sensing performance, we use classification accuracy as the primary metric, evaluated for both the legitimate Rx and the attacker. Communication performance is measured by the packet loss rate (PLR). As a baseline, we include a scenario without a metasurface where the Tx uses an omnidirectional antenna. All experiments strictly comply with the Institutional Review Board guidelines of our institute.

VI. Evaluation Result

In this section, we first examine the effectiveness of the proposed beamforming algorithm. We then evaluate CloakFi in terms of privacy protection, sensing accuracy, and communication performance, followed by an investigation of key system parameters.

A. Beamforming Performance of Metasurface

We visualize the beam pattern generated by the first metasurface row in Fig. 7. The top panel shows the signal amplitudes of the two designed beamforming vectors across different angles, while the bottom panel presents their corresponding phase differences. The remaining metasurface rows exhibit similar patterns and are thus omitted for brevity. Notably, both the communication direction ($-20°$) and the sensing direction ($50°$) exhibit strong beamforming gains, confirming that the design supports high-quality communication and sensing. More importantly, in the communication direction, the two beamforming vectors produce similar magnitudes and phases, ensuring that random switching between them does not degrade communication performance. Meanwhile, in the

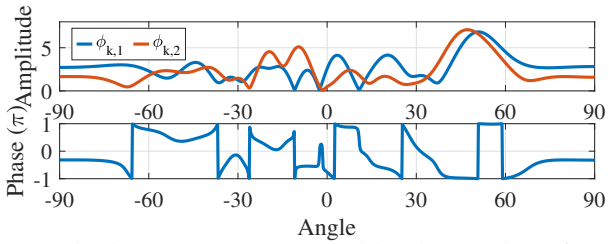


Fig. 7: The beam pattern generated by the two beamforming vectors for the first row of the metasurface.

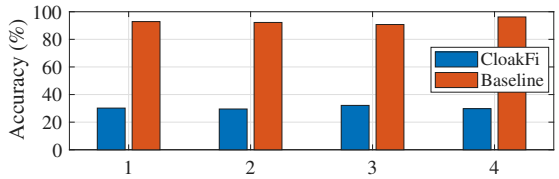


Fig. 8: Privacy protection with different locations.

	PP	SL	UD	IS	DZ	CL	WA	DS	DC
PP	0.88	0	0	0	0	0.08	0	0	0.04
SL	0.03	0.87	0.03	0	0	0.03	0	0.03	0
UD	0	0	1	0	0	0	0	0	0
IS	0	0	0	1	0	0	0	0	0
DZ	0	0	0	0	1	0	0	0	0
CL	0	0	0	0	0	1	0	0	0
WA	0	0	0	0	0	0	0.96	0.04	0
DS	0	0	0	0	0	0	0	0.88	0.13
DC	0	0	0	0	0	0	0	0.09	0.91

(a) CloakFi.

	PP	SL	UD	IS	DZ	CL	WA	DS	DC
PP	0.86	0	0.03	0	0.03	0.09	0	0	0
SL	0.02	0.91	0.04	0	0	0.02	0	0	0
UD	0	0	1	0	0	0	0	0	0
IS	0	0	0	1	0	0	0	0	0
DZ	0	0	0	0	1	0	0	0	0
CL	0	0	0	0	0	1	0	0	0
WA	0	0	0	0	0	0	0.91	0.09	0
DS	0	0	0	0	0	0	0.04	0.92	0.04
DC	0	0	0	0	0	0	0.03	0.16	0.81

(b) Baseline.

Fig. 9: The accuracy of the legitimate sensing Rx.

sensing direction, the magnitudes remain similar, but the phase difference fluctuates around $\pm\pi$, inducing significant CSI variation. This variation serves to obfuscate the CSI measured at the attacker and enhance privacy protection.

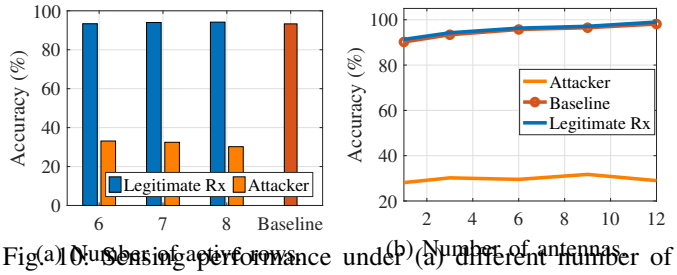
B. Overall Performance

Privacy protection. We first evaluate the privacy-preserving capability of CloakFi. Fig. 8 shows the attacker's sensing accuracy at four different locations. In the baseline scenario without any protection, the attacker achieves an average recognition accuracy of approximately 93 %. After applying CloakFi, the accuracy drops significantly to around 30 %, representing a reduction of over 60 % and highlighting the strong privacy protection capability of our approach. The underlying reason is that, regardless of the attacker's location, successfully inferring the target's gesture requires capturing the signal transmitted by the Tx and subsequently reflected or scattered by both the metasurface and the sensing target. Consequently, the resulting CSI inevitably includes both target-related information and artificial variations deliberately introduced by the metasurface in the sensing direction. Crucially, these two components, target-relevant signals and metasurface-induced perturbations, are deeply intertwined and indistinguishable from each other in the observed CSI. This makes it fundamentally difficult for an attacker to isolate a meaningful gesture.

Sensing performance. To evaluate sensing of legitimate Rx, Fig. 9 presents the accuracy of the legitimate Rx under both the proposed CloakFi and a baseline setup without the metasurface. In the baseline scenario, the average gesture recognition accuracy reaches 93.3 %. With CloakFi, the accuracy is slightly higher at 94.2 %, suggesting that the masking and demasking method is effective and does not degrade sensing performance. The marginal improvement mainly stems from the additional gain provided by the beamforming of the metasurface. The beamforming enhances signal power in the sensing direction, improving the effective sensing SNR. It is worth noting that the improvement appears marginal, mainly because the baseline accuracy is already very high.

TABLE I: PLR (%) with different MCS indices.

MCS index	MCS 3	MCS 5	MCS 7
CloakFi	1.07%	3.90%	17.78%
Baseline	1.27%	13.78%	40.56%



(a) Number of active rows. (b) Number of antennas.

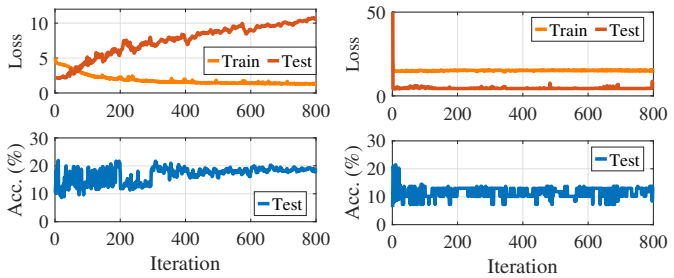
Fig. 10: Sensing performance under (a) different number of active metasurface rows and (b) different number of antennas.

Communication performance. Tab. I compares the PLR of the legitimate communication Rx in CloakFi with that of the baseline under varying MCS indices. CloakFi consistently outperforms the baseline, particularly at higher MCS levels where SNR requirements are more stringent. This improvement is not solely due to the energy focusing capability of the metasurface, but more importantly, stems from our beamforming design, which ensures that the two beamforming vectors produce highly similar signal amplitudes and phases in the communication direction. This design mitigates the negative impact of random vector switching, thereby preserving stable and reliable transmission.

C. Impact of Parameters and Extended Experiments

In our system, masking is realized by randomly selecting one of two beamforming vectors for each row of the metasurface. The effectiveness of this masking naturally depends on the number of active rows. To assess this, we vary the number of activated rows and measure the sensing performance of both the legitimate Rx and the attacker, as shown in Fig. 10(a). As illustrated, the legitimate Rx achieves slightly higher sensing accuracy as more rows are activated, owing to improved beamforming that concentrates power toward the sensing direction and boosts SNR. In contrast, the attacker's accuracy steadily decreases since additional rows introduce greater randomness into the CSI, thereby strengthening the obfuscation effect and hindering meaningful information extraction.

Increasing the number of antennas generally enhances spatial sensing resolution. To examine whether this also benefits the attacker, we simultaneously expand the antenna array at both the legitimate sensing Rx and the attacker by aggregating CSI data from multiple Wi-Fi NICs and merging them post-capture. Fig. 10(b) presents sensing accuracy as the antenna



(a) Classifier in SignFi. (b) ResNet.

Fig. 11: Training loss and testing loss/accuracy under (a) the classifier adopted in SignFi and (b) ResNet.

count grows from 1 to 12. As expected, both the legitimate Rx and the baseline attacker (without defense) achieve higher accuracy with more antennas, eventually reaching a plateau due to spatial diversity. In contrast, under the protection of CloakFi, the attacker's accuracy shows little improvement even with more antennas. This stems from the metasurface-induced spatial perturbations that entangle sensing information with artificial noise, as analyzed in Section IV. Such coupling prevents the attacker from separating useful features, regardless of antenna count. These results confirm that CloakFi remains effective even when the attacker scales up antenna resources.

In prior experiments, both the attacker and legitimate sensing Rx relied on the same classifier trained on decrypted CSI. A natural question arises: is the attacker's poor performance simply due to model–data mismatch? To test a stronger adversary, we consider an extreme case where the attacker independently collects masked CSI (e.g., via long-term eavesdropping) and trains a classifier from scratch. As shown in Fig. 11(a), with the SignFi model, the training loss decreases but the test loss rises, and accuracy remains below 20 %, indicating failure to extract meaningful features from encrypted data. To exclude the possibility of insufficient model capacity, we further adopt a ResNet-based classifier. As Fig. 11(b) illustrates, even with greater depth, losses converge to a non-trivial bound and test accuracy stays consistently low. This result arises because, without the shared key, the attacker only sees randomized CSI sequences for the same gesture. Such randomness breaks the temporal and spatial consistency required for representation learning, preventing reliable model training. Overall, these findings demonstrate the robustness and privacy-preserving strength of CloakFi, even against enhanced adversarial strategies.

VII. Related Work

CloakFi is closely related to Wi-Fi sensing and ISAC, as well as privacy-preserving techniques for wireless sensing.

Wi-Fi sensing and ISAC. Commodity Wi-Fi devices have become a widely adopted platform for contactless human sensing, benefiting from their device-free, non-intrusive nature, insensitivity to lighting conditions, and ubiquity. They have enabled a variety of applications, such as localization [7], [30], [31], activity recognition [32]–[35], identity authentication [36], [37], and vital sign monitoring [38], [39]. However, these works primarily focus on sensing, requiring the transmission of dedicated sensing packets, thereby separating sensing from communication. With the rise of ISAC, researchers have explored ways to achieve ISAC directly over Wi-Fi. One approach leverages the beamforming feedback, which is transmitted in plaintext [3], [40]. Yet, its low sampling rate limits fine-grained sensing. Another line of work extracts CSI from regular data packets, but such CSI is distorted by communication-oriented beamforming. To address this, SenCom [9] introduces a passive signal processing solution, while VersaBeam [41] actively exploits beamforming to achieve high-quality ISAC. Despite these advances, an important aspect has been overlooked: the publicly accessible CSI embedded in Wi-Fi packets can also result in privacy leakage. Our work tackles this missing piece.

Privacy preservation. Existing privacy-preserving approaches in wireless sensing can be broadly categorized into two groups. The first line of work leverages the Tx's capability to obfuscate channel information. For example, MIMOCrypt [19] and Secur-Fi [42] encrypt CSI through precoding shared between trusted transceivers, while other efforts [43]–[46] attempt to modify pilots or inject adversarial perturbations to the transmitted signals to confuse sensing models. However, these techniques either degrade communication performance or impose stringent requirements on hardware and protocol modifications, which limit their applicability to low-cost IoT platforms. The second category relies on external mechanisms. PhyCloak [20] introduces relays to distort physical-layer information but requires expensive full-duplex hardware. IRShield [23], on the other hand, leverages metasurfaces to obfuscate CSI, yet it overlooks the impact on legitimate users' ISAC performance. In light of these limitations, there remains a gap: how to safeguard privacy while maintaining ISAC capabilities on low-cost devices. This work aims to address this gap.

VIII. Conclusion

In this paper, we presented CloakFi, a practical metasurface-enabled system that achieves high-quality communication and sensing while preserving user privacy. Our design introduces a novel metasurface beamforming design that generates two distinct beamforming vectors per metasurface row, maximizing signal variation in the sensing direction while maintaining stable, high-gain transmission in the communication direction. By switching between these vectors, CloakFi introduces artificial perturbations that effectively obscure sensitive sensing information. To guarantee legitimate sensing, we further develop a time-domain masking and demasking method, allowing only authorized Rx to identify and extract valid sensing information. Experiment results with commodity Wi-Fi devices demonstrate that CloakFi achieves strong privacy protection while preserving high ISAC performance, highlighting its practicality for diverse ISAC scenarios.

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